

Name: _____

Section I (6 marks)

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Attempt Questions 1 to 6

Allow about 12 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 6

1. What is the gradient of any line which is perpendicular to $5x + 2y = 3$?

- (A) $\frac{2}{5}$ (B) $\frac{-2}{5}$ (C) $\frac{5}{2}$ (D) $\frac{-5}{2}$

2. Expand and simplify $(3 - \sqrt{7})^2$

- (A) $(16 - 2\sqrt{21})$ (B) 16 (C) $(16 - 6\sqrt{7})$ (D) 2

3. Simplify the expression $\sec x \sin x$

- (A) $\operatorname{cosec} x$ (B) $\cos x$ (C) $\tan x$ (D) 1

4.

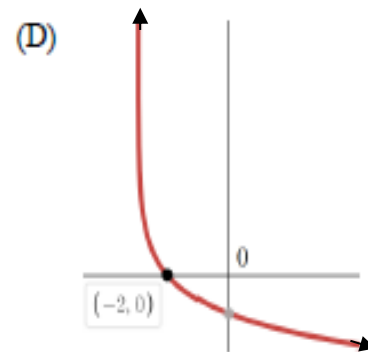
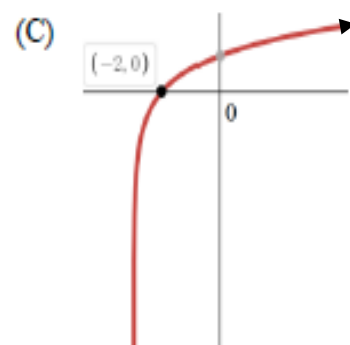
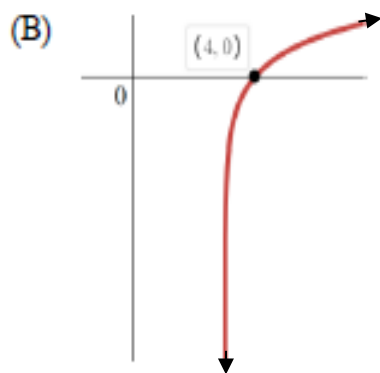
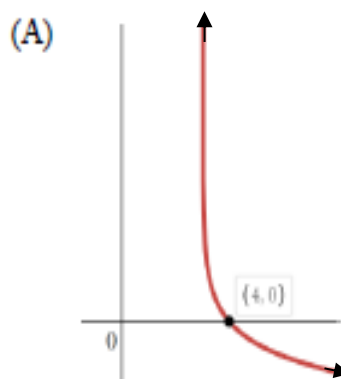
Calculate the size of angle α in the triangle, correct to the nearest second



- (A) $48^\circ 35' 25.36''$ (B) $48^\circ 35'$ (C) $48^\circ 35' 25''$ (D) $48^\circ 35' 26''$

5.

Which of the following graphs refer to $y = -\log_e(x-3)$.



6.

Given $y = \frac{-1}{2x^2}$ find y'

(A) $\frac{4}{x^3}$

(B) $\frac{1}{x^3}$

(C) $\frac{1}{x}$

(D) $\frac{-1}{x^3}$

End of Section I

Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 7 (10 marks)

Marks

1

(a)

Solve: $-\frac{2x+1}{3} = 5$

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(b)

Use the quadratic formula to solve $x^2 + 12x - 9 = 0$, giving your answer in simplest surd form.

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Question 7 continues on the following page

(c)

Find the centre and radius of the circle with equation:

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$$x^2 - 6x + y^2 + 2y - 6 = 0$$

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Question 7 continues on the following page

(d) (i) Solve the equation: $|x-2|=4$ 1

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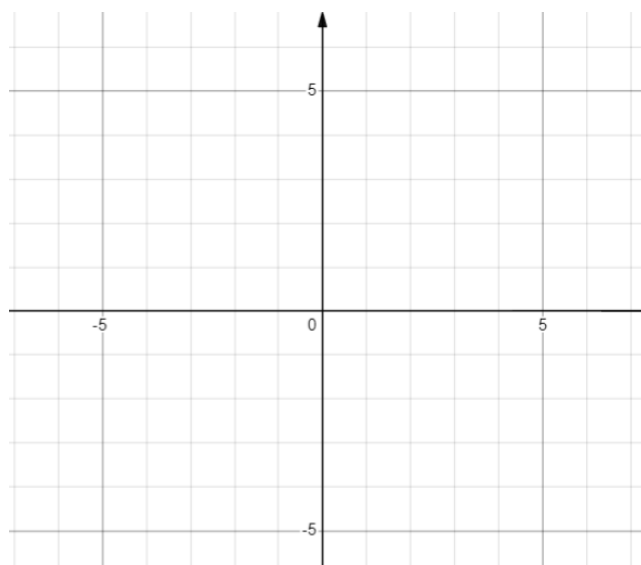
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(ii) Draw a sketch of $y=|x-2|$ and $y=4$ on the same set of axes, clearly indicating the coordinates of any point of intersection and intercepts. 2



(iii) Hence or otherwise solve the inequality $|x-2|\geq 4$ 2

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End of Question 7

If you use this space, clearly indicate which question you are answering.

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Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 8 (15 marks)

Marks
2

- (a) Consider the function given below:

$$f(x) = \begin{cases} x^2 - 9 & \text{when } x \leq 0 \\ -9 + x & \text{when } x > 0 \end{cases}$$

Evaluate $f(2) + f(0) - f(-2)$

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- (b) Given $f(x) = \frac{3}{x}$ and $g(x) = x - 2$

2

State the domain and range for $f(g(x))$

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Question 8 continues on the following page

(c)

Calculate the exact value of:

2

$$(\cos 315^\circ + \tan 135^\circ)(\sin 45^\circ - \tan 135^\circ)$$

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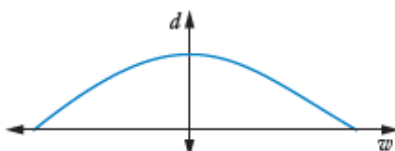
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(d)

A satellite dish has the shape of a parabola shown below, with equation

$$d = \frac{-w^2}{800} + 200 \quad \text{where } d \text{ is the depth of the dish in metres.}$$



(i)

Show, algebraically, that the quadratic function is even.

1

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Question 8 continues on the following page

(ii) Find the maximum depth of the dish from the top of the span. **1**

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(iii) What is the total width of the satellite dish? **2**

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(iv) Calculate the depth of the dish at a point 100 metres from its endpoint. **2**

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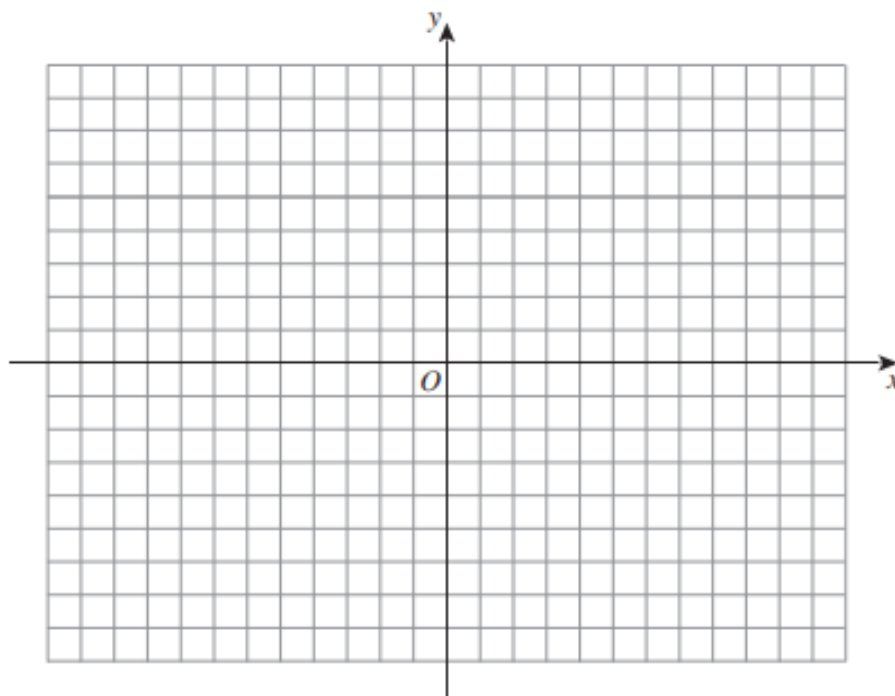
Question 8 continues on the following page

(e)

Sketch the graph of $y = e^x + 1$

3

Label the features of your graph clearly.



Extra writing space

If you use this space, clearly indicate which question you are answering.

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End of Question 8

Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

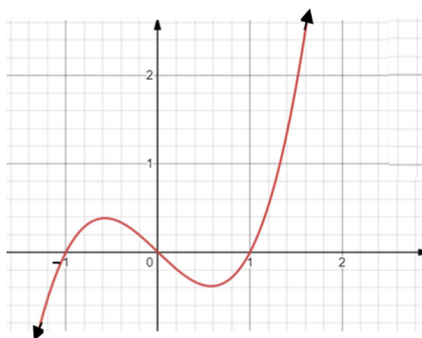
Section II - Answer in the space provided

Question 9 (15 marks)

Marks

- (a) (i) Write the equation of the monic polynomial shown below, in expanded form.

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- (ii) The graph in part (i) is shifted 1 unit to the left. Show that this new function is:

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$$g(x) = x^3 + 3x^2 + 2x$$

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Question 9 continues on the following page

- (b) (i) Find the equation of the tangents to the curve $y = x^2 - 4x$, at the points where the curve cuts the x-axis. 2

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- (ii) Hence, what is the point of intersection of these tangents? 1

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- (c) Prove $\frac{\tan \theta}{\sec \theta - 1} - \frac{\tan \theta}{\sec \theta + 1} = 2 \cot \theta$ 2

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Question 9 continues on the following page

(d) Find the derivative of the function $f(x) = 2x^2 - 3x$ from first principles. 3

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(e) Solve $2\cos 2x = 1$ for $0 \leq x \leq \pi$. 3

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End of Question 9

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Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 10 (15 marks)

Marks

- (a) Give the exact value of 195° in radians

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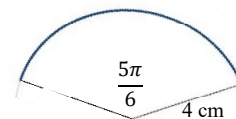
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- (b) (i) XY is an arc of a circle with centre O.
 The radius $OX = 4$ cm and angle $XOY = \frac{5\pi}{6}$ radians.
 Find the length of the arc XY correct to 1 decimal place.

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- (ii) The arc XY is extended to point Z forming the new sector XOZ.
 The sector XOZ has an area of $\frac{26\pi}{3} \text{ cm}^2$
 Find angle XOZ subtended at the centre, in radians.

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Question 10 continues on the following page

(c) Solve $\cos^2 x - \cos x = 0$ for $[0, 2\pi]$ 3

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(d) Differentiate:

(i) $f(x) = 5x(x^3 + 1)$ 1

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(ii) $y = \frac{4x - 3}{3x + 2}$ 2

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(iii) $y = 6x^2 e^{-x}$ 2

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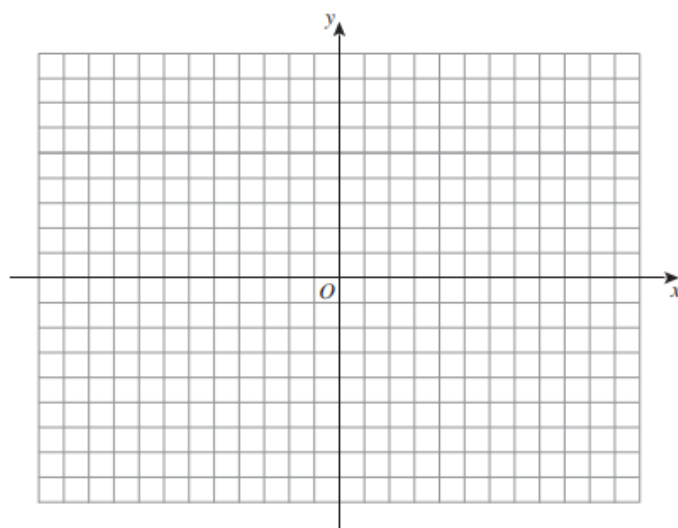
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Question 10 continues on the following page

(e)

Use the axes below to show graphically and give the number of solutions of the equation $\log_e x - x = -2$

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End of Question 10

Extra writing space

If you use this space, clearly indicate which question you are answering.

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Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 11 (15 marks)

Marks

2

- (a) Simplify, using the log laws,
 $2 \log_2 25 + \log_2 16 - \log_2 20$

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- (b) Solve for x , correct to 1 decimal place
 $e^{2x} = 200$

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- (c) Find $\frac{d}{dx} \left[(4e^{3x} - 1)^8 \right]$

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Question 11 continues on the following page

(d) If $R = t^2 - 12t + 2$

(i) Find the average rate of change of R between $t = 1$ and $t = 4$ 1

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(ii) What is the instantaneous rate of change of R at $t = 4$ 1

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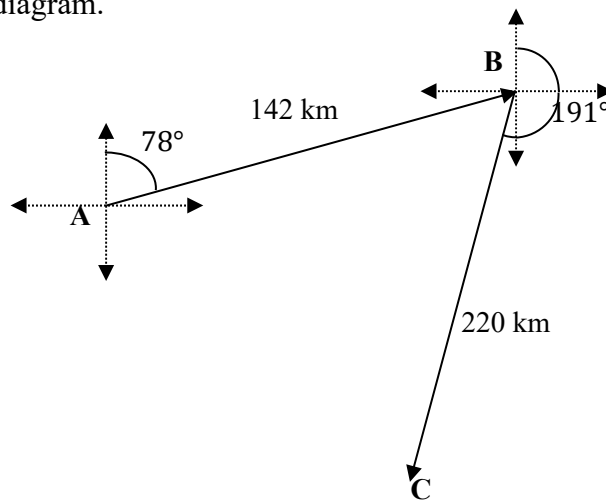
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(e) Angie leaves island A in a boat and sails 142 km on a bearing of 078° , to island B. She then sails on a bearing of 191° for 220 km to island C, as shown in the diagram.



**Diagram not
to scale**

(i) Show that $\angle ABC = 67^\circ$ 2

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- (ii) Calculate the distance from island C to island A, to the nearest km. 2

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- (iii) If Angie wants to sail from island C directly back to island A on what bearing should Angie sail? 3
(Answer to the nearest degree)

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End of Question 11

Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 12 (14 marks)

Marks

(a)

Use the Chain Rule to differentiate:

$$y = \sqrt{5-x}$$

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(b)

The displacement, x metres, of an object moving along a straight line over time, t seconds, is given by

$$x = (t^3 + 1)^4$$

(i)

Find the initial displacement and velocity.

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(ii)

Find its acceleration after 2 seconds (answer in scientific notation, correct to 3 significant figures).

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Question 12 continues on the following page

- (iii) Show that the particle is never at the origin. 2

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- (c) The amount A grams of a given carbon isotope in a dead tree trunk is given by

$$A = A_0 e^{-kt}$$

where A_0 is the original amount of isotope in the dead tree, k is a positive constant, and where the time t is measured in years from the death of the tree.

- (i) Show that A satisfies the equation $\frac{dA}{dt} = -kA$ 1

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- (ii) The amount of isotope is halved every 5500 years. 3
Show that the value of k is 1.26027×10^{-4}

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Question 12 continues on the following page

- (c)
(iii) For a particular dead tree trunk, the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die? 2
Answer to the nearest 1000 years.

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End of paper

If you use this space, clearly indicate which question you are answering.

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Name: _____

Teacher: _____

Fort Street High School 2022

Preliminary HSC Assessment Task 3



Mathematics Advanced

Time allowed: 2 hours (plus 10 minutes reading time)

General Instructions

- Marks may be deducted for careless or untidy work
- Write using blue or black pen
- NESA approved calculators may be used
- A formulae sheet is provided with this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks – 90

Section I: Pages 1 to 2 6 marks

- Attempt questions 1-6, using the answer sheet, page 26, provided
- Allow about 10 minutes for this section

Section II: Pages 3 to 26 84 marks

- Attempt questions 7-12, using the space provided in the booklet
- Allow about 1 hour and 50 minutes for this section

Outcome	Assessment Area Description and Marking Guidelines
MA11-1	Uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems
MA11-2	Uses the concepts of functions and relations to model, analyse and solve practical problems
MA11-3	Uses the concepts and techniques of Trigonometry in the solution of equations and problem solving involving geometric shapes
MA11-4	Uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities
MA11-5	Interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems
MA11-6	Manipulates and solves expressions using the logarithmic and index laws, and uses logarithms and exponential functions to solve practical problems
MA11-9	Provides reasoning to support conclusions which are appropriate to the context

Final Marks

Question	Mark
Section I	
Multiple Choice	
Questions 1-6	/6
Section II	
Question 7	/10
Question 8	/15
Question 9	/15
Question 10	/15
Question 11	/15
Question 12	/14
Total	/90
	%

2022 Year 11 Advanced Final Examination
Mathematics Advanced
Section I – Multiple Choice Answer Sheet

(Answers)

Name _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Allow about 10 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word correct and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☒ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☒ D ☐
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☐ C ☒ D ☐
5. A ☒ B ☐ C ☐ D ☐
6. A ☐ B ☒ C ☐ D ☐

Name: _____ (Answers)

Section I (6 marks)

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Attempt Questions 1 to 6

Allow about 12 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 6

1. What is the gradient of any line which is perpendicular to $5x + 2y = 3$?
- $2y = -5x + 3$
 $y = -\frac{5}{2}x + \frac{3}{2}$

- (A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{5}{2}$ (D) $-\frac{5}{2}$

2. Expand and simplify $(3 - \sqrt{7})^2 = (3 - \sqrt{7})(3 - \sqrt{7}) = 16 - 6\sqrt{7}$

- (A) $(16 - 2\sqrt{21})$ (B) 16 (C) $(16 - 6\sqrt{7})$ (D) 2

3. Simplify the expression $\sec x \sin x = \frac{1}{\cos x} \cdot \sin x = \frac{\sin x}{\cos x}$

- (A) $\cos \sec x$ (B) $\cos x$ (C) $\tan x$ (D) 1

4.

Calculate the size of angle α in the triangle, correct to the nearest second



$$\tan \alpha = \frac{3}{\sqrt{7}}$$

$$\alpha = \tan^{-1} \left(\frac{3}{\sqrt{7}} \right)$$

$$= 48^\circ 35' 25.36''$$

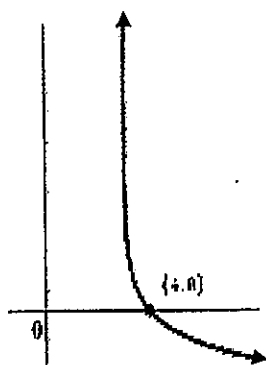
$$\approx 48^\circ 35' 25''$$

- (A) $48^\circ 35' 25.36''$ (B) $48^\circ 35'$ (C) $48^\circ 35' 25''$ (D) $48^\circ 35' 26''$

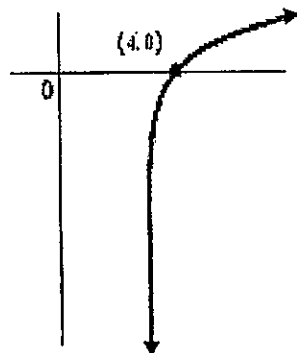
5.

Which of the following graphs refer to $y = -\log_e(x-3)$.
translated 3 units to the right
reflection in x-axis

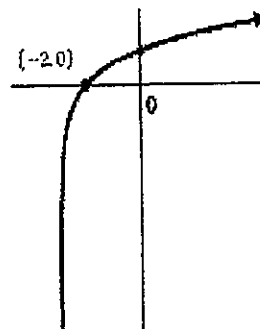
(A)



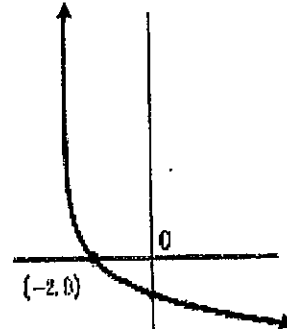
(B)



(C)



(D)



6.

Given $y = \frac{-1}{2x^2}$ find y'

$$y = -\frac{1}{2} \cdot x^{-2}, \quad y' = x^{-3} = \frac{1}{x^3}$$

(A) $\frac{4}{x^3}$

(B) $\frac{1}{x^3}$

(C) $\frac{1}{x}$

(D) $\frac{-1}{x^3}$

End of Section I

Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 7 (10 marks)

Marks
1

(a)

Solve: $-\frac{2x+1}{3} = 5$

$$2x+1 = -15$$

$$2x = -16$$

$$x = -8$$



(b)

Use the quadratic formula to solve $x^2 + 12x - 9 = 0$, giving your answer in simplest surd form.

2

$$x = \frac{-12 \pm \sqrt{12^2 - 4 \times 1 \times -9}}{2(1)}$$

$$= \frac{-12 \pm \sqrt{180}}{2}$$

$$\begin{aligned} &\sqrt{180} \\ &= \sqrt{36 \times 5} \\ &= 6\sqrt{5} \end{aligned}$$

$$= \frac{-12 \pm 6\sqrt{5}}{2}$$

$$= \frac{2(-6 \pm 3\sqrt{5})}{2}$$

$$= -6 \pm 3\sqrt{5}$$



Question 7 continues on the following page

(c)

Find the centre and radius of the circle with equation:

2

$$x^2 - 6x + y^2 + 2y - 6 = 0$$

$$x^2 - 6x + 9 + y^2 + 2y + 1 = 6 + 10$$

$$(x-3)^2 + (y+1)^2 = 16$$

$$\therefore \text{radius} = 4, \text{centre } (3, -1)$$

/ working

/ answer including
(both radius + centre)

Question 7 continues on the following page

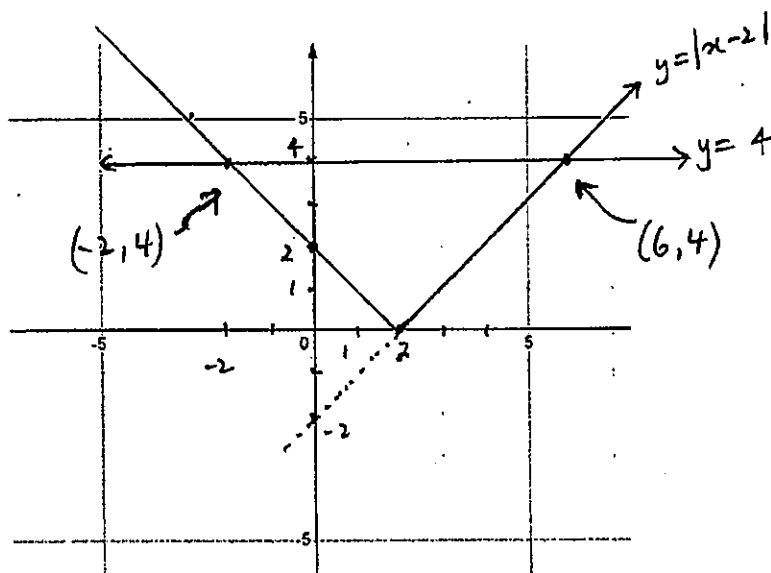
(d) (i) Solve the equation: $|x-2|=4$

1

$$\begin{aligned}x-2 &= 4 & , & & x-2 &= -4 \\ \underline{x &= 6} & , & & \underline{x &= -2}\end{aligned}$$

(ii) Draw a sketch of $y=|x-2|$ and $y=4$ on the same set of axes, clearly indicating the coordinates of any point of intersection and intercepts.

2



✓ pts of int.
and
intercepts
✓ graphs/sketch

(iii) Hence or otherwise solve the inequality $|x-2| \geq 4$

2

$$\begin{aligned}x-2 &\geq 4 & , & & x-2 &\leq -4 \\ \underline{x &\geq 6} & , & & \underline{x &\leq -2}\end{aligned}$$

End of Question 7

Name: _____

Teacher	Fentoullis	Hussein	Razzaghi	Kaur	Moon	Wilkinson
	☐	☐	☐	☐	☐	☐

Section II - Answer in the space provided

Question 8 (15 marks)

Marks
2

- (a) Consider the function given below:

$$f(x) = \begin{cases} x^2 - 9 & \text{when } x \leq 0 \\ -9 + x & \text{when } x > 0 \end{cases}$$

Evaluate $f(2) + f(0) - f(-2)$

$$\begin{aligned} &= (-9 + 2) + (0^2 - 9) - [(2)^2 - 9] \\ &= -7 - 9 - (-5) \\ &= -11 \end{aligned}$$

- (b) Given $f(x) = \frac{3}{x}$ and $g(x) = x - 2$

2

State the domain and range for $f(g(x))$

$$f(g(x)) = \frac{3}{x-2}$$

$$\text{Domain: } (-\infty, 2) \cup (2, \infty)$$

$$\text{Range: } (-\infty, 0) \cup (0, \infty)$$

Question 8 continues on the following page

(c)

Calculate the exact value of:

2

$$(\cos 315^\circ + \tan 135^\circ)(\sin 45^\circ - \tan 135^\circ)$$

$$= \left[\frac{1}{\sqrt{2}} + (-1) \right] \left[\frac{1}{\sqrt{2}} - (-1) \right]$$

$$= \left(\frac{1}{\sqrt{2}} - 1 \right) \left(\frac{1}{\sqrt{2}} + 1 \right) \quad \checkmark$$

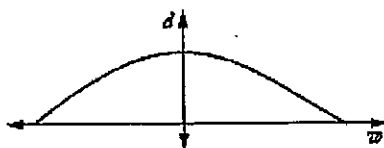
$$= \frac{1}{2} - 1$$

$$= -\frac{1}{2} \quad \checkmark$$

(d)

A satellite dish has the shape of a parabola shown below, with equation

$$d = \frac{-w^2}{800} + 200$$

where d is the depth of the dish in metres.

(i)

Show, algebraically, that the quadratic function is even.

1

$$f(w) = \frac{-w^2}{800} + 200$$

$$f(-w) = \frac{-(-w)^2}{800} + 200$$

$$= \frac{-w^2}{800} + 200$$

$$= f(w)$$

$\therefore$ since $f(w) = f(-w)$
the function is even $\checkmark$

Question 8 continues on the following page

- (ii) Find the maximum depth of the dish from the top of the span.

1

$$\text{When } w=0, \quad d = -\frac{(0)^2}{800} + 200$$

$$= 200$$

$\therefore$ the maximum depth is 200 m

- (iii) What is the total width of the satellite dish?

2

$$\text{When } d=0, \quad 0 = -\frac{w^2}{800} + 200$$

$$\frac{w^2}{800} = 200$$

$$w^2 = 160\,000$$

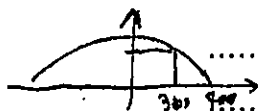
$$\therefore w = \pm \sqrt{160\,000}$$

$$= \pm 400$$

$$\therefore \text{the total width} = 400 + 400 \\ = 800 \text{ m}$$

- (iv) Calculate the depth of the dish at a point 100 metres from its endpoint.

2



$$\text{when } w=300$$

$$d = -\frac{(300)^2}{800} + 200$$

$$= -\frac{90\,000}{800} + 200$$

$$= 87.5 \text{ m}$$

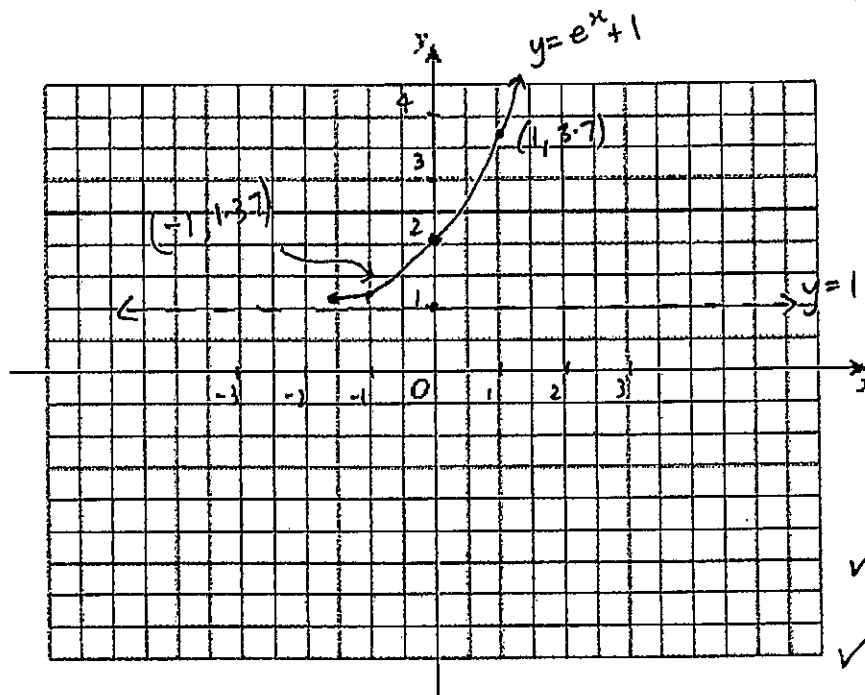
Question 8 continues on the following page

(e)

Sketch the graph of $y = e^x + 1$

3

Label the features of your graph clearly.



✓ y intercept / shape
✓ asymptote
✓ additional point/s on the curve.

Extra writing space

If you use this space, clearly indicate which question you are answering.

when $x = -1$ when $x = 1$

$y = e^{-1} + 1$ $y = e + 1$

$= \frac{1}{e} + 1$ $= 3.7$

$= 1.37$

$= 1.368$

End of Question 8

Name: _____

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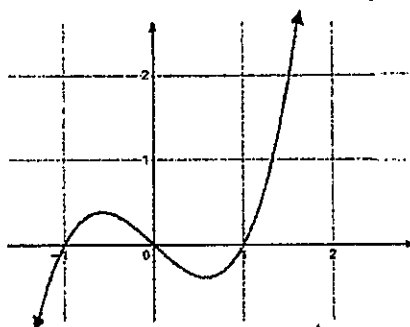
Section II - Answer in the space provided

Question 9 (15 marks)

Marks

- (a) (i) Write the equation of the monic polynomial shown below, in expanded form.

2



Using zeros: $y = (x+1)(x)(x-1)$ ✓

$$= x(x^2 - 1)$$

$$y = x^3 - x$$
 ✓

- (ii) The graph in part (i) is shifted 1 unit to the left. Show that this new function is:

2

$$g(x) = x^3 + 3x^2 + 2x$$

Using

zeros: -2, -1, 0

$(x+2)$

$(x+1)$

x

$$g(x) = x(x+1)(x+2)$$
 ✓

$$= x(x^2 + 3x + 2)$$

$$= x^3 + 3x^2 + 2x$$
 ✓

Question 9 continues on the following page

(b) (i)

Find the equation of the tangents to the curve $y = x^2 - 4x$, at the points where the curve cuts the x-axis.

2

Curve cuts the x-axis when $y = 0$
 $0 = x(x - 4)$
at $x = 0, 4$

$m_T = \frac{dy}{dx} = 2x - 4$
at $x = 0, m_T = 2(0) - 4 = -4$
at $x = 4, m_T = 2(4) - 4 = 4$

$\therefore (0, 0) \text{ and } (4, 0)$

Eqn of tangent, $y - 0 = -4(x - 0)$
 $y = -4x$

Eqn of tangent, $y - 0 = 4(x - 4)$
 $y = 4x - 16$

(ii) Hence, what is the point of intersection of these tangents?

1

points of int:

$y = -4x$ — (1)
 $y = 4x - 16$ — (2)

(1) + (2)

$2y = -16$
 $y = -8$

When $y = -8, -8 = -4(0) \quad x = 2$

pt. of intersection is $(2, -8)$

(c)

Prove $\frac{\tan \theta}{\sec \theta - 1} - \frac{\tan \theta}{\sec \theta + 1} = 2 \cot \theta$

2

LHS = $\frac{\tan \theta (\sec \theta + 1) - \tan \theta (\sec \theta - 1)}{\sec^2 \theta - 1}$

= $\frac{\tan \theta \sec \theta + \tan \theta - \tan \theta \sec \theta + \tan \theta}{\tan^2 \theta}$

= $\frac{2 \tan \theta}{\tan^2 \theta} = \frac{2}{\tan \theta} = 2 \cot \theta = \text{RHS}$

Question 9 continues on the following page

(d)

Find the derivative of the function $f(x) = 2x^2 - 3x$ from first principles.

3

$$\begin{aligned}
 f(x) &= 2x^2 - 3x & f'(x) &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 3x - 3h - (2x^2 - 3x)}{h} \\
 f(x+h) &= 2(x+h)^2 - 3(x+h) & &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h} \\
 &= 2(x^2 + 2xh + h^2) - 3x - 3h & &= \lim_{h \rightarrow 0} \frac{4x + 2h - 3}{1} \\
 &= 2x^2 + 4xh + 2h^2 - 3x - 3h & &= 4x - 3
 \end{aligned}$$

(e)

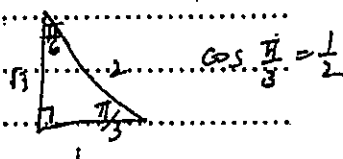
Solve $2\cos 2x = 1$ for $0 \leq x \leq \pi$

3

$$\cos 2x = \frac{1}{2} \quad \text{for } 0 \leq x \leq \frac{\pi}{2} \text{ then}$$

$$0 \leq 2x \leq \pi$$

$$2x \text{ acute/reference } \frac{\pi}{3}$$



$$2x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\checkmark \checkmark \text{ (answers.)}$$

End of Question 9

Name: _____

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Section II - Answer in the space provided

Question 10 (15 marks)

Marks

- (a) Give the exact value of 195° in radians

1

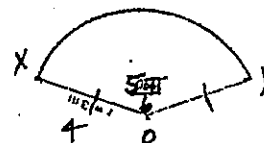
$$195 \times \frac{\pi}{180} = \frac{13\pi}{12} \quad \checkmark$$

- (b) (i) XY is an arc of a circle with centre O.

2

The radius $OX = 4$ cm and angle $XOY = \frac{5\pi}{6}$ radians.

Find the length of the arc XY correct to 1 decimal place.



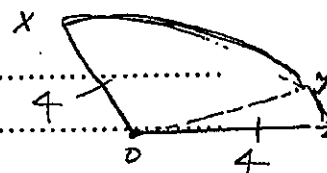
$$\begin{aligned} L &= r\theta \\ &= 4 \times \frac{5\pi}{6} \\ &= \frac{10\pi}{3} \quad \checkmark \\ &= 10.4719... \approx 10.5 \text{ cm} \quad \checkmark \end{aligned}$$

- (ii) The arc XY is extended to point Z forming the new sector XOZ.

2

The sector XOZ has an area of $\frac{26\pi}{3} \text{ cm}^2$

Find angle XOZ subtended at the centre, in radians.



$$\begin{aligned} A &= \frac{1}{2} r^2 \theta \\ \frac{26\pi}{3} &= \frac{1}{2} (16) \theta \quad \checkmark \\ \frac{52\pi}{3} &= 16\theta \\ \theta &= \frac{52\pi}{3} \times \frac{1}{16} \quad \checkmark \\ \theta &= \frac{13\pi}{12} \quad \checkmark \end{aligned}$$

Question 10 continues on the following page

(c)

Solve $\cos^2 x - \cos x = 0$ for $[0, 2\pi]$

3

$$\cos x (\cos x - 1) = 0$$

$$\cos x = 0, \quad \cos x - 1 = 0$$

$$\cos x = 1$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, x = 0, 2\pi$$

(d)

Differentiate:

(i)

$$f(x) = 5x(x^3 + 1)$$

1

$$f(x) = 5x^4 + 5x$$

(OR)

$$f'(x) = 20x^3 + 5$$

$$\begin{aligned} f'(x) &= 5x \cdot 3x^2 + (x^3 + 1) \cdot 5 \\ &= 15x^3 + 5x^3 + 5 \\ &= 20x^3 + 5 \end{aligned}$$

(ii)

$$y = \frac{4x-3}{3x+2}$$

2

$$\text{Let } u = 4x-3$$

$$v = 3x+2$$

$$u' = 4$$

$$v' = 3$$

$$y' = \frac{u'v + v'u}{v^2} = \frac{12x+8-12x+9}{(3x+2)^2}$$

$$= \frac{17}{(3x+2)^2}$$

(iii)

$$y = 6x^2 e^{-x}$$

2

$$\text{Let } u = 6x^2$$

$$v = e^{-x}$$

$$u' = 12x$$

$$v' = -e^{-x}$$

$$y' = uv' + v u' = (6x^2)(-e^{-x}) + (12x)(e^{-x})$$

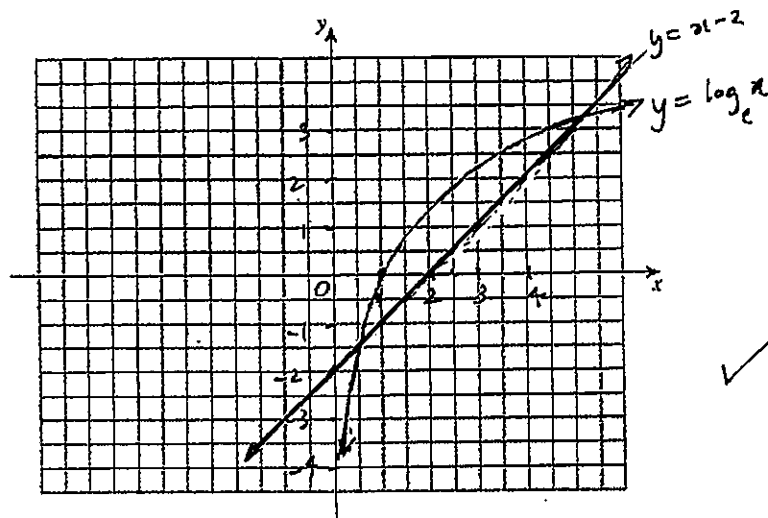
$$= 6x^2 e^{-2} (2-x) \quad \text{(OR)} \quad -6x e^{-x} (x-2)$$

Question 10 continues on the following page.

(e)

Use the axes below to show graphically, the number of solutions of the equation $\log_e x - x = -2$

2



$\log_e x = x - 2$
.....
.....
 $y = \log_e x$
.....
 $y = x - 2$
.....
.....
 $\therefore 2 \text{ solutions}$ ✓

End of Question 10

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Section II - Answer in the space provided

Question 11 (15 marks)

Marks

- (a) Simplify, using the log laws,
 $2\log_2 25 + \log_2 16 - \log_2 20$

2

$$\begin{aligned}
 &= 2\log_2 5^2 + \log_2 2^4 - \log_2 (4 \times 5) \\
 &= 4\log_2 5 + 4\log_2 2 - [2\log_2 2 + \log_2 5] \\
 &= \underline{3\log_2 5 + 2}
 \end{aligned}$$

- (b) Solve for x , correct to 1 decimal place

2

$$e^{2x} = 200$$

$$\begin{aligned}
 \ln e^{2x} &= \ln 200 \\
 2x \ln e &= \ln 200 &= 2.649... \\
 2x &= \ln 200 &\div 2 \\
 x &= \frac{\ln 200}{2} &= \underline{2.6}
 \end{aligned}$$

- (c) Find $\frac{d}{dx}[(4e^{3x} - 1)^8]$

2

$$\begin{aligned}
 &= 8(4e^{3x} - 1)^7 \cdot 4e^{3x} \cdot 3 \\
 &= \underline{96(4e^{3x} - 1)^7 e^{3x}}
 \end{aligned}$$

Question 11 continues on the following page

(d) If $R = t^2 - 12t + 2$

- (i) Find the average rate of change of R between $t=1$ and $t=4$

1

$R = t^2 - 12t + 2$

$f(1) = 1^2 - 12(1) + 2 = -9$

$f(4) = 4^2 - 12(4) + 2 = -30$

$\therefore \text{Rate of Change} = \frac{-30 - (-9)}{4 - 1} = \frac{-21}{3} = -7$

- (ii) What is the instantaneous rate of change of R at $t=4$

1

$f'(t) = 2t - 12$

$= 2(4) - 12$

$= 8 - 12$

$= -4$

- (e) Angie leaves island A in a boat and sails 142 km on a bearing of 078° , to island B. She then sails on a bearing of 191° for 220 km to island C, as shown in the diagram.

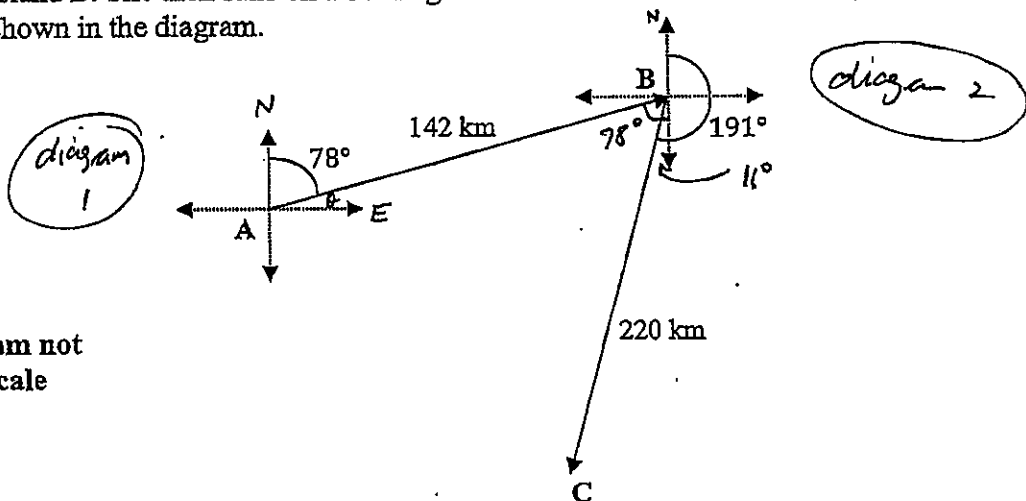


Diagram not to scale

- (i) Show that $\angle ABC = 67^\circ$

2

In diagram 1: $\theta = 90 - 78 = 12^\circ$

In diagram 2: alternate $\angle$ s $\angle ABC = 78^\circ$

$\therefore \angle ABC = \dots$

$= 78 - (191 - 180)$

$= 67^\circ$

- (ii) Calculate the distance from island C to island A, to the nearest km.

2

$$d_{AC} = \sqrt{142^2 + 220^2 - 2 \times 142 \times 220 \cos 67^\circ}$$

$$= \sqrt{44\,151.119 \dots}$$

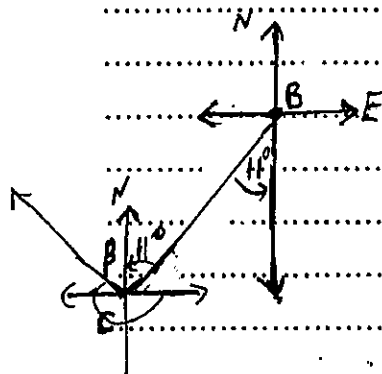
$$= 210.12$$

$$\approx 210 \text{ km}$$

- (iii) Angie wants to sail from island C directly back to island A. Given the bearing of B from C is 011° , on what bearing should Angie sail? (Answer to the nearest degree)

3

Since $191 - 180 = 11^\circ$
(the bearing of B from C)



$\angle ACB:$

$$\cos \angle ACB = \frac{210^2 + 220^2 - 142^2}{2 \times 210 \times 220}$$

$$= 0.782956 \dots$$

$$\angle ACB = \cos^{-1}(\dots)$$

$$= 38.467 \dots$$

$$= 38^\circ$$

$$\beta = 38 - 11$$

$$= 27^\circ$$

$\therefore$ The bearing of A from C

$$360 - 27$$

$$= 333^\circ$$

End of Question 11

Name: _____

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Section II - Answer in the space provided

Question 12 (14 marks)

Marks

(a)

Use the Chain Rule to differentiate:

2

$$y = \sqrt{5-x}$$

Let $u = 5-x$

$\therefore y = \sqrt{u} = u^{1/2}$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ (OR) $y = (5-x)^{1/2}$

$\frac{du}{dx} = -1$

$\frac{dy}{du} = \frac{1}{2} u^{-1/2} = \frac{1}{2\sqrt{u}}$

$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{u}} \cdot (-1) = -\frac{1}{2\sqrt{5-x}}$

$\frac{dy}{dx} = -\frac{1}{2\sqrt{5-x}}$

(b)

The displacement, x metres, of an object moving along a straight line over time, t seconds, is given by

$$x = (t^3 + 1)^4$$

(i)

Find the initial displacement and velocity.

2

$x(0) = (0^3 + 1)^4 = 1 \text{ m}$ ✓

$\dot{x} = 4(t^3 + 1)^3 \cdot 3t^2$

$= 12t^2(t^3 + 1)^3$

$\dot{x}(0) = 12(0)(0^3 + 1)^3 = 0 \text{ ms}^{-1}$ ✓

(ii)

Find its acceleration after 2 seconds (answer in scientific notation, correct to 3 significant figures).

2

$\ddot{x} = 12t^2 [3(t^3 + 1)^2 \cdot 3t^2] + (t^3 + 1)^3 \cdot 24t$

$= 108t^4 [(t^3 + 1)^2] + 24t(t^3 + 1)^3$

$\therefore \ddot{x}(2) = 108(2)^4 [(2^3 + 1)^2] + 24(2)(2^3 + 1)^3$

$= 174960$

$\div 175000 \text{ ms}^{-2} = 1.75 \times 10^5 \text{ (3 sig. fig.)}$

Question 12 continues on the following page

- (iii) Show that the particle is never at the origin.

2

$$x = (t^3 + 1)^4$$

$$\text{for } x=0, \quad t^3 + 1 = 0$$

$$t^3 = -1$$

as $t \geq 0$ $t^3 \geq 0 \therefore$ Never at the origin.

no sol.

- (c) The amount A grams of a given carbon isotope in a dead tree trunk is given by

$$A = A_0 e^{-kt}$$

where A_0 and k are positive constants, and where the time t is measured in years from the death of the tree.

- (i) Show that A satisfies the equation $\frac{dA}{dt} = -kA$

1

$$\frac{dA}{dt} = A_0 \times -k \times e^{-kt}$$

$$= -k A_0 e^{-kt}$$

$$= -k(A) \quad (\text{since } A_0 e^{-kt} = A)$$

- (ii) The amount of isotope is halved every 5500 years.

3

Show that the value of k is 1.26027×10^{-4}

$$\text{When } t = 5500, \quad A = \frac{1}{2} A_0$$

$$\therefore \frac{1}{2} A_0 = A_0 e^{-5500k}$$

$$\ln \frac{1}{2} A_0 = \ln A_0 e^{-5500k}$$

$$\ln \frac{1}{2} = -5500k$$

$$\therefore k = \frac{\ln \frac{1}{2}}{-5500} = 0.0001260276 \dots$$

$$= 1.260276 \times 10^{-4}$$

Question 12 continues on the following page

- (iii) For a particular dead tree trunk, the amount of isotope is only 15% of the original amount in the living tree. How long ago did the tree die?
Answer to the nearest 1000 years.

2

$$A = 0.15 A_0$$

$$\therefore 0.15 A_0 = A_0 e^{-kt}$$

$$0.15 = e^{-kt} \quad \checkmark$$

$$\ln 0.15 = \ln e^{-kt}$$

$$= -kt / \ln e$$

$$\therefore t = \frac{\ln 0.15}{-k} \quad \checkmark$$

$$= 15053.3107 \dots$$

$$= \underline{15\,000 \text{ years}} \quad (\text{nearest } 1000 \text{ yr})$$

End of paper